Light

GCSE Computer Science - Cyber Security Prevention (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is egress filtering on a firewall?

- ○ **A.** Monitoring and controlling outgoing traffic leaving the network
- ○ **B.** Filtering emails before they are sent
- ○ **C.** Blocking all incoming connections
- ○ **D.** Encrypting data before storage

Q2 (1 mark)

What is the purpose of email scanning in anti-malware software?

- ○ **A.** To make emails load faster

- ○ **B.** To check email attachments and links for malicious content before users open them
- ○ **C.** To automatically reply to emails
- ○ **D.** To change the font of emails

Q3 (1 mark)

What is account deprovisioning?

- ○ **A.** Creating new user accounts
- ○ **B.** Increasing account privileges
- ○ **C.** Removing access rights and disabling accounts when users leave or change roles
- ○ **D.** Making passwords more complex

Q4 (1 mark)

What is the purpose of a CAPTCHA on a login page?

- ○ **A.** To verify that the user is human and not an automated bot attempting mass login attempts
- ○ **B.** To make the page more colourful
- ○ **C.** To encrypt the password
- ○ **D.** To store the password securely

Q5 (1 mark)

Why do organisations require visitors to sign in and wear badges?

- ○ **A.** To make visitors feel welcome
- ○ **B.** To advertise the company name
- ○ **C.** To track who is on the premises, identify unauthorised individuals, and create a record of visitor access
- ○ **D.** To collect email addresses for marketing

Q6 (1 mark)

What is end-to-end encryption?

- ○ **A.** Encryption that only works at the end of the month
- ○ **B.** Encryption of the first and last characters only
- ○ **C.** Encryption that ends after 24 hours
- ○ **D.** Encryption where only the sender and recipient can read the message - not even the service provider

Q7 (1 mark)

What is network segmentation?

- ○ **A.** Dividing a network into smaller, isolated sections to contain breaches and control traffic flow
- ○ **B.** Cutting network cables into segments

- ○ **C.** Increasing network speed
- ○ **D.** Connecting all devices to one network for simplicity

Q8 (1 mark)

What is application whitelisting?

- ○ **A.** Making application windows white for easier reading
- ○ **B.** Only allowing approved applications to run - blocking all others by default
- ○ **C.** Listing applications that should be deleted
- ○ **D.** Colouring dangerous applications white

Q9 (1 mark)

What is the purpose of security awareness training for employees?

- ○ **A.** To teach employees to hack other companies
- ○ **B.** To replace all technical security controls
- ○ **C.** To educate staff about security threats, safe practices, and their role in protecting company data
- ○ **D.** To punish employees who make mistakes

Q10 (1 mark)

Why do security experts recommend enabling automatic software updates?

- ○ **A.** They ensure security patches are applied promptly without relying on users to remember
- ○ **B.** Automatic updates make software more colourful
- ○ **C.** Automatic updates are always free
- ○ **D.** They make computers run faster

Q11 (1 mark)

In penetration testing, what is a grey box test?

- ○ **A.** Testing conducted in a grey-painted room
- ○ **B.** Testing only grey-coloured equipment
- ○ **C.** A test that always produces inconclusive results
- ○ **D.** A test where testers have partial knowledge of the target system - between black box (no knowledge) and white box (full knowledge)

Q12 (1 mark)

What is a clean desk policy?

- ○ **A.** A rule about cleaning desks with disinfectant
- ○ **B.** A policy allowing only white desks in the office
- ○ **C.** A requirement to clear sensitive documents and lock away materials when leaving the workspace

- ○ **D.** A rule about eating lunch at desks

Q13 (2 marks)

A company encrypts all customer data but stores the encryption keys in a plain text file on the same server.
What security problem does this create?

- ○ **A.** No problem - the encryption is still effective
- ○ **B.** Anyone accessing the server can read the keys and decrypt all data, making the encryption useless; keys must be stored separately and securely
- ○ **C.** The encryption becomes too slow
- ○ **D.** The data becomes permanently unreadable

Q14 (2 marks)

An IT department uses a single shared administrator account that all technicians know the password for.
What security risks does this create?

- ○ **A.** No individual accountability for actions; the password is known by many increasing exposure risk; when staff leave, the password must change for everyone
- ○ **B.** No risks - sharing is efficient
- ○ **C.** The account becomes more secure with more users
- ○ **D.** Shared accounts are required by law

Q15 (2 marks)

A firewall is configured with a 'default deny' policy - blocking all traffic except what is explicitly permitted.
What is the security advantage of this approach?

- ○ **A.** It makes the firewall easier to configure
- ○ **B.** It speeds up all network traffic
- ○ **C.** Users can access any website they want
- ○ **D.** Only specifically approved traffic passes; new or unknown threats are blocked by default; this is more secure than allowing everything except known threats

Q16 (2 marks)

Modern browsers include security features like blocking known malicious websites and warning about dangerous downloads.
How do these features protect users?

- ○ **A.** They make web pages load faster
- ○ **B.** They replace the need for anti-malware software entirely
- ○ **C.** They provide a first line of defence by stopping users from visiting known phishing sites or downloading identified malware before damage occurs
- ○ **D.** They only work when the computer is offline

Q17 (2 marks)

Before conducting a penetration test, the security team and client agree on a detailed scope document.
Why is defining scope important?

- ○ **A.** Scope defines what systems can be tested, testing methods allowed, timing constraints, and boundaries - preventing testers from accidentally disrupting critical systems or exceeding legal authorisation
- ○ **B.** Scope documents are only for legal decoration
- ○ **C.** Scope makes the test take longer than necessary
- ○ **D.** Scope limits are unnecessary if testers are experienced

Q18 (2 marks)

Many websites allow password recovery using security questions like 'What is your mother's maiden name?' or 'What city were you born in?' What security weakness do these questions have?

- ○ **A.** The questions are too difficult to answer
- ○ **B.** Answers can often be found through social media research or public records, allowing attackers to reset passwords without knowing the original
- ○ **C.** Security questions make passwords stronger
- ○ **D.** They are more secure than passwords

Q19 (2 marks)

An organisation disables USB ports on all workstations and provides secure, company-approved alternatives for file transfer.
What threats does this address?

- ○ **A.** It makes computers look neater with fewer cables
- ○ **B.** USB ports generate too much heat
- ○ **C.** It speeds up the computer by freeing resources

○ **D.** It prevents malware introduction via infected USB drives and blocks unauthorised data copying onto removable media

Q20 (2 marks)

An organisation installs an Intrusion Detection System (IDS) on their network.
What is the difference between an IDS and a firewall?

○ **A.** They are exactly the same thing with different names

○ **B.** An IDS is physical hardware; a firewall is software only

○ **C.** A firewall blocks traffic based on rules; an IDS monitors traffic for suspicious patterns and alerts administrators to potential attacks but typically does not block automatically

○ **D.** Firewalls detect intrusions; IDS blocks traffic

Q21 (2 marks)

A laptop used by a travelling employee has full disk encryption enabled. What protection does this provide if the laptop is stolen?

○ **A.** All data on the drive is encrypted; without the decryption key, the thief cannot access any files even if they remove the hard drive and connect it to another computer

○ **B.** The thief cannot sell the laptop

○ **C.** The laptop will automatically return to the owner

○ **D.** Full disk encryption makes laptops theft-proof

Q22 (2 marks)

A contractor needs access to company systems for a two-week project. The IT department creates an account that automatically expires after the project end date.
Why is expiring access beneficial?

- ○ **A.** Expiring access is just for decoration in the system
- ○ **B.** Access is automatically revoked without relying on manual deprovisioning; this prevents forgotten accounts from becoming security vulnerabilities
- ○ **C.** The contractor works faster with expiring access
- ○ **D.** Expiring accounts cost less to create

Q23 (2 marks)

An organisation requires all employees to sign an Acceptable Use Policy (AUP) covering computer and network use.
What is the purpose of this policy?

- ○ **A.** It is purely legal protection with no practical benefit
- ○ **B.** It automatically prevents all policy violations
- ○ **C.** It defines what users may and may not do with company systems, sets expectations, and provides grounds for disciplinary action if rules are violated
- ○ **D.** It replaces the need for technical security controls

Q24 (3 marks)

A hospital network is segmented into zones: patient records, medical devices, guest Wi-Fi, and administrative systems. Each zone has controlled connections to others.

Analyse how this architecture protects patient safety and data.

- ○ **A.** Segmentation has no effect on security - all zones are equally vulnerable
- ○ **B.** Segmentation makes the network too slow for medical use
- ○ **C.** All zones should be combined for simpler management
- ○ **D.** Each zone is isolated: compromised guest Wi-Fi cannot reach medical devices; infected admin computers cannot access patient records directly; critical systems are protected from less-secure zones; attackers must breach multiple barriers

Q25 (3 marks)

A company conducts annual security awareness training but continues to see employees falling for phishing attacks. They consider switching to monthly micro-training with simulated phishing tests.
Analyse why the new approach might be more effective.

- ○ **A.** Monthly training wastes more time than annual training
- ○ **B.** Annual training is forgotten between sessions; monthly micro-training reinforces concepts regularly; simulated phishing provides practical experience and identifies who needs additional help; continuous learning builds lasting security habits
- ○ **C.** The frequency of training does not affect security outcomes
- ○ **D.** Employees should be punished for failing tests instead of receiving more training

Q26 (3 marks)

A company upgrades from a basic packet-filtering firewall to an application-layer firewall (also called a next-generation firewall).
Analyse what additional protection this provides.

- ○ **A.** Application-layer firewalls are identical to packet filters with a different name
- ○ **B.** Application-layer firewalls only protect applications, not networks
- ○ **C.** Basic packet filters provide better security than application-layer inspection
- ○ **D.** Packet filters only examine headers (addresses and ports); application-layer firewalls inspect actual content, can identify specific applications regardless of port, detect malware in traffic, and apply rules based on application behaviour - providing deeper protection

Q27 (3 marks)

Security researchers find that some malware now detects when it is running in a sandbox or virtual machine and remains dormant to avoid analysis.
Analyse the implications for anti-malware protection.

- ○ **A.** Sandboxing remains perfect protection against all malware
- ○ **B.** Sandbox-aware malware evades behavioural analysis by not executing malicious code during testing; this cat-and-mouse evolution means security requires multiple detection methods - signature, behaviour, sandbox, machine learning - as no single approach catches everything
- ○ **C.** Virtual machines should never be used for security
- ○ **D.** This only affects researchers, not real-world protection

Q28 (3 marks)

An organisation adopts a 'zero trust' security model with the principle 'never trust, always verify' - requiring authentication and authorisation for every access request, even from inside the network.

Analyse why this approach addresses modern threats.

- ○ **A.** Zero trust is unnecessary if the network has a strong perimeter firewall
- ○ **B.** Zero trust means nobody can access any systems
- ○ **C.** Trusting internal users improves productivity without security risks
- ○ **D.** Traditional perimeter security assumes internal users are trusted, but attackers who breach the perimeter or compromise internal accounts face no further barriers; zero trust verifies every request regardless of source, limiting lateral movement and containing breaches

Q29 (3 marks)

A penetration test identifies critical vulnerabilities. The organisation patches them and requests a retest. The retest finds the original vulnerabilities are fixed but discovers new issues introduced by the patches.
Analyse what this reveals about the security improvement process.

- ○ **A.** Retesting is unnecessary if patches are applied
- ○ **B.** Fixes can introduce new vulnerabilities; retesting verifies remediation effectiveness and catches unintended consequences; security is iterative - each improvement should be validated; the testing and fixing cycle continues until acceptable risk levels are achieved
- ○ **C.** Finding new issues means the patches failed completely
- ○ **D.** Penetration testing should only be done once

Q30 (3 marks)

A security audit reveals: excellent technical controls (firewalls, encryption, access controls), but weak physical security (unlocked server

room), untrained staff (clicking phishing links), and no incident response plan.
Analyse why comprehensive security requires addressing all areas.

- ○ **A.** Strong technical controls are sufficient - other areas do not matter
- ○ **B.** Physical security is outdated in modern computing
- ○ **C.** Staff training is unnecessary if technical controls are strong
- ○ **D.** Attackers target the weakest point: physical access bypasses network controls; untrained staff bypass technical defences; lack of response planning extends damage duration - comprehensive security requires technical, physical, human, and procedural elements working together

Total: **55** marks • Time: **30 min**

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